

REPORT

INTRODUCTION

Collective cell migration concerns the movement of a group of cells in response to environmental or chemical cues, and is an emergent property of many interacting individual cells. Collective migration can happen when the volume of cells is enforced, but not too strongly so (λ_{Volume} parameter). Cells further back can place new pixels inside a cell in the front on its backside, making the front cell 'under volume' and making it place new pixels on its other side more quickly. Through this interaction, the cell in the back can push the front cell to move more quickly than it otherwise would have. By using the Cellular Potts Model, we can explore how individual cells interacting with their environment give rise to such collective behavior.

Obstacles represent some environmental constraints that affect collective cell migration by guiding the movement of cells or disrupting it. By varying the number of obstacles, changes in the collective cell migration can be observed. Investigating these effects helps reveal the underlying mechanisms that guide cell and group behaviors. This experiment aims to explore how obstacles change the collective migration behaviour of cells by observing simulations with 4, 9, 16, or 25 obstacles.

METHODS

The Artistoo framework was used to implement the two-dimensional Cellular Potts Model simulations. The model was defined on a square grid of 200x200 pixels with wrapped dimensions. In addition to the background, there were two types of cells in the simulations: moving cells and obstacles. Obstacles were implemented by creating a new type of cell with different parameters in order to distinguish the two types of cells and to give them desired properties.

Obstacles were initialized with a target volume (V) of 150 pixels and volume constraint strength (λ_V) of 500 to ensure they are static and do not deform. Their perimeter (P) was set to 10 with a strong perimeter constraint strength (λ_P) 200, keeping their shape strongly constrained. To further prevent any movement, their act (λ_{act} , Max_{act}) was set to 0. Additionally, the adhesion between obstacles and moving cells ($J_{\text{obstacle-cell}}$) was set to a 1000 to ensure moving cells do not stick to obstacles or penetrate them.

On the other hand, moving cells had completely different parameters that would allow them to move and explore their environment, reproducing collective cell migration. These parameters were directly taken from the self-study exercise 1.3. In total, there were 37 moving cells on the grid. 37 was chosen because it is the maximum number of cells that allowed all obstacle configurations without breaking the simulation. Each moving cell was initialized with V of 200 pixels, making them (very roughly) twice the size of the obstacles,

and λ_V of 50 to allow them moderate compressibility. Their P was set to 180 and λ_p to 2 to allow them to take on irregular shapes during their migration. Their movement, and the active migration, was controlled using activity constraints with λ_{act} of 200 and Max_{act} 20. These parameters allowed the reproduction of collective migration.

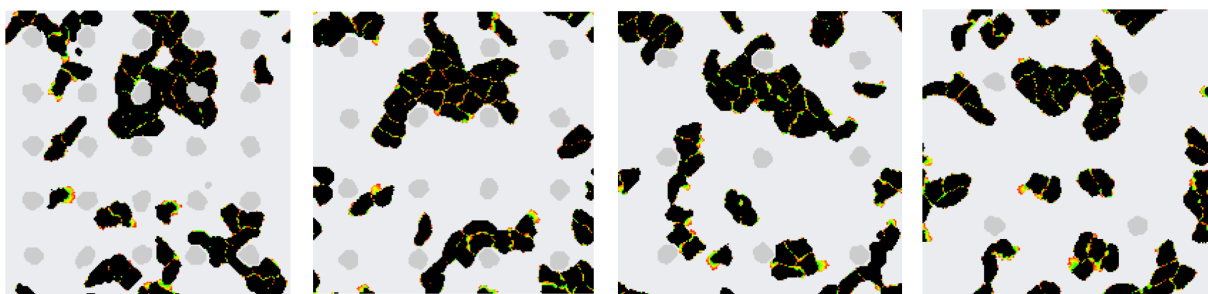
There were 4 experiments in total. All were performed under the same conditions and with identical parameters, with the exception of the number of obstacles. In order to investigate how obstacles change the collective migration behavior of cells, 4 obstacle configurations were explored: 4, 9, 16, and 25. Obstacles placed on the lattice were equally spaced from each other, forming a grid-like structure. Additionally, throughout the experiments all simulations had temperature (T) 20, which controls the stochasticity of the system, allowing cells to deform and migrate. Each simulation was initialized by placing the obstacles first, and then randomly seeding moving cells. Each simulation ran for 600 Monte Carlo steps, which took about 10 minutes in wall-clock time at 1fps, and the observations and results are reported in the Results section.

To get quantitative data, we run a python script on the outputs that calculates the average speed and distance traversed as well as those of the fastest and slowest cells, along with the standard deviation in speeds. This allows us to analyse the cell movements for both environments with and without obstacles in a quantitative, objective way and also base our qualitative analysis on a factual basis.

RESULTS

Initial states

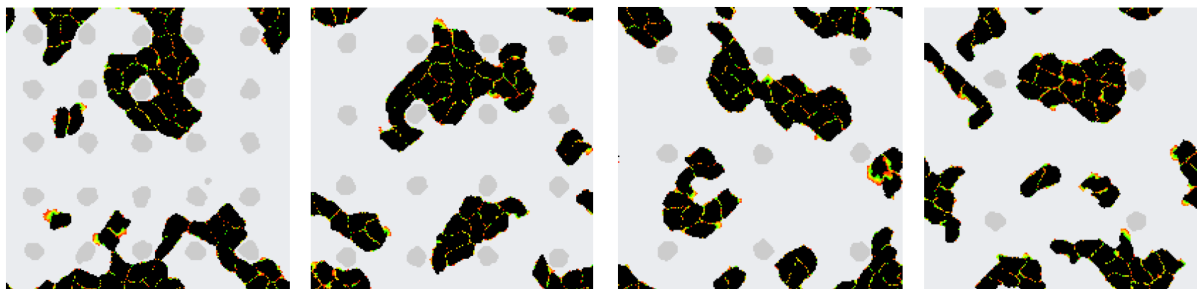
Here, we initialised the state. The images are taken after 500 warmup steps, which is why cells have already flocked together a bit. Not much to see yet!



Initial state: 25 obstacles. Initial state: 16 obstacles. Initial state: 9 obstacles. Initial state: 4 obstacles.

End states after 1 hour

While the environments with more obstacles form bigger mega-clusters, the ones with less obstacles form more, but smaller clusters.



End state: 25 obstacles.

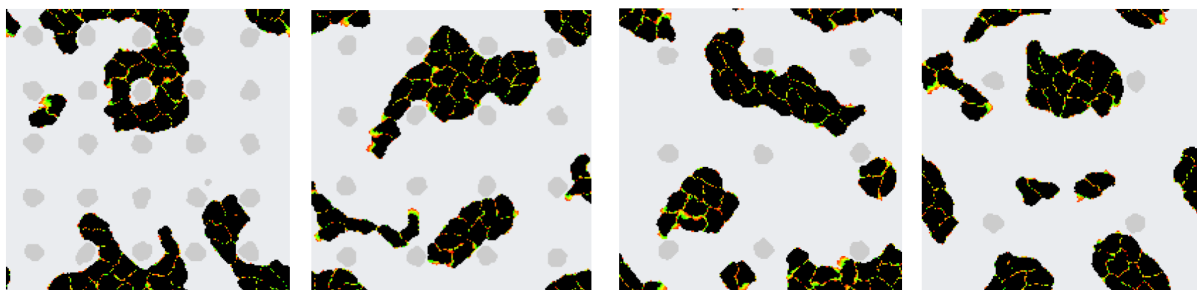
End state: 16 obstacles.

End state: 9 obstacles.

End state: 4 obstacles.

End states after 2 hours

It is apparent that the environments with 4 obstacles leave the cells enough space so that they can form stable clusters apart from each other. There is not much migrating behaviour apart from finding a group and sticking to it.



25 obstacles

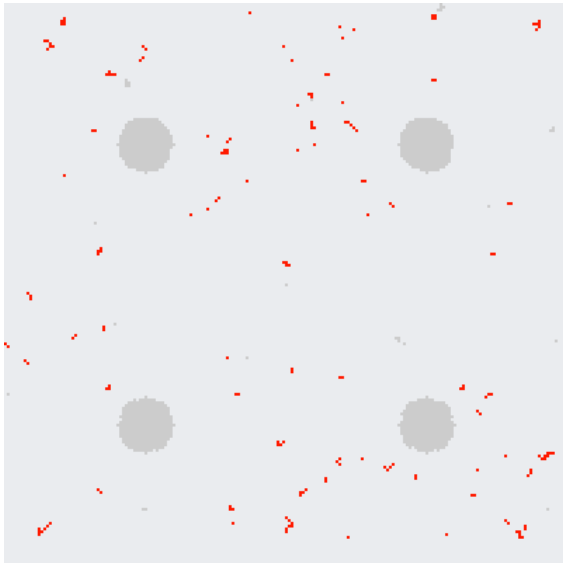
16 obstacles

9 obstacles

4 obstacles

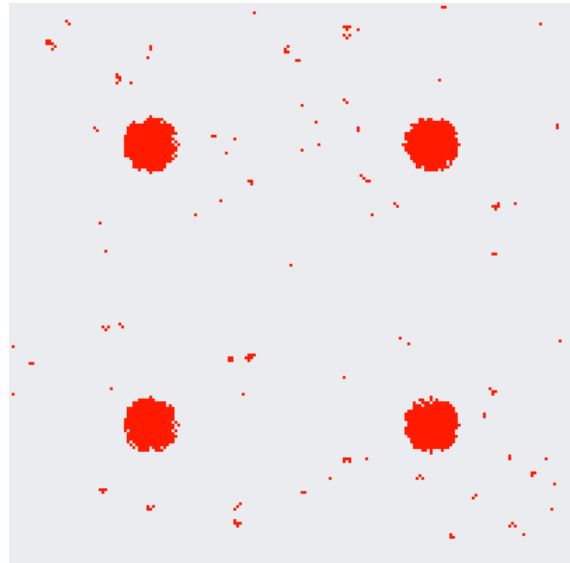
Video & Speed Comparison

Without obstacles:



[Click to view animated version on Giphy](#)

With 16 obstacles:



[Click to view animated version on Giphy](#)

Comparison:

Dataset: 4440 observations

Step range: 0 - 590, representing 1 minute at 10 fps

Number of moving cells: 74

Speed: pixels/timestep	Without obstacles	With 16 obstacles
Average speed across all cells	0.0960	0.0599
Std dev of cell speeds	0.1928	0.0789
Fastest cell avg speed	1.4168	0.3903
Slowest cell avg speed	0.0112	0.0078

Total pixels travelled	Without obstacles	With obstacles
Average distance per cell	56.63	35.34
Furthest travelling cell	835.92	230.25
Least travelling cell	6.58	4.59

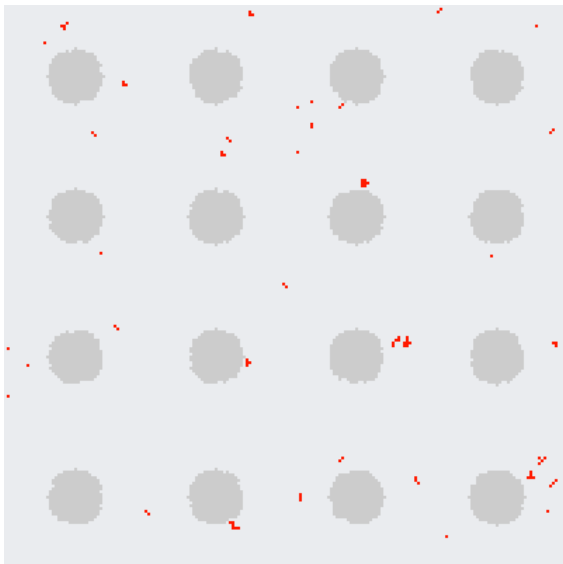
In our setup, when there are no obstacles, the cells travel further on average. The cells move slower with obstacles but either way, they don't really move/migrate much at all. They mostly

move around for a while until they find another cell and then stay fairly stationary and stick together. More obstacles means that there is less free space and that they don't have to search as long to find other cells to cluster to, therefore reducing their total mobility.

Perhaps with parameters that would incentivize them to move more and stick to each other less we would get more group migrating behaviour. We have also observed that with some code adjustments, cells actually travel further with obstacles than without, and much more in both cases.

Revised Video & Speed Comparison

Without obstacles (revised):



[Click to view animated version on Giphy](#)

With obstacles (revised):



[Click to view animated version on Giphy](#)

We will further explore why this happens or if this is perhaps the intended behaviour in the next submission after the draft.

Revised Comparison:

Dataset: 2220 observations

Step range: 0 - 590, representing 1 minute at 10 fps

Number of moving cells: 37

Speed: pixels/timestep	Without obstacles	With obstacles
Average speed across all cells	0.1359	0.2228
Std dev of cell speeds	0.2003	0.2953

Fastest cell avg speed	1.0964	1.0708
Slowest cell avg speed	0.0321	0.0302

Total pixels travelled	Without obstacles	With obstacles
Average distance per cell	80.18	131.44
Furthest travelling cell	646.89	631.77
Least travelling cell	18.95	17.85

DISCUSSION

There are several limitations. First, the parameters used were taken from the self-study exercise 1.3 and no other parameter configurations were explored. This is the main reason why cells form groups and then stop migrating. Furthermore, this study only considered 4 obstacle configurations. Moreover, the obstacles were always equally spaced in a grid-like pattern, meaning that the environment did not change much between the 4 obstacle variations.

Second, there are bugs in the code and the implementation of this experiment is imperfect. This may cause cells to behave weirdly, not like intended. Although the obstacles were round, they were not circular as intended. While this may have very little effect on the cell behaviors, it could have potentially led to undesired behaviors. Is it therefore recommended that future studies explore irregularly placed obstacles as well as dynamic obstacles, and it is also advised to check the code for any unintended bugs.